

OpenClaw AI Voice Interaction Configuration

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1. Course Overview

This course will introduce how to configure the AI voice interaction pipeline in OpenClaw, including the location of voice configuration files, descriptions of key parameters, startup methods, and troubleshooting for common issues.

After completing this course, you will be able to:

- Know which configuration file is used for AI voice interaction
- Understand the meaning of key parameters in `.env`
- Master the method to switch between Chinese and English voice recognition
- Correctly start the voice program

2. Overall Voice Interaction Pipeline

The current voice interaction flow in OpenClaw can be simply understood as:

1. Microphone captures voice

2. Program performs silence detection and segmentation
3. ASR engine converts voice to text
4. Recognition results are published to OpenClaw
5. OpenClaw returns results
6. [optional] TTS broadcasts the returned results (you can choose whether to broadcast as replies can be very long)

The core configuration file for this pipeline is:

```
/home/sunrise/voice_to_openclaw/.env
```

3. Voice Configuration File Description

3.1 Configuration File Location

Default uses the accompanying factory image

```
/home/sunrise/voice_to_openclaw/.env
```

If this file does not exist, you can copy it from the template first:

```
cd /home/sunrise/voice_to_openclaw
cp .env.example .env
```

3.2 Minimum Configuration to Check Before Startup

It is recommended to confirm at least the following items:

- ASR_ENGINE
- DASHSCOPE_API_KEY
- DASHSCOPE_LANGUAGE or XUNFEI_ASR_LANGUAGE
- DEVICE
- SPEAKER_DEVICE
- OPENCLAW_CONFIG_PATH
- ENABLE_SERIAL_WAKE

4. Key Parameter Description

4.1 Voice Recognition Engine

```
ASR_ENGINE=dashscope
```

Currently supports three modes:

- dashscope: Tongyi Bailian native voice recognition interface, suitable for direct online use
- xunfei_ws: iFlytek WebSocket voice recognition, suitable for overseas version configuration
- faster_whisper: Local Whisper recognition, suitable for local operation (not supported by RDK)

4.2 Microphone and Speaker Devices

```
DEVICE=0  
SPEAKER_DEVICE=0
```

These two parameters determine which audio input and output devices the program uses.

Suitable scenarios for modification:

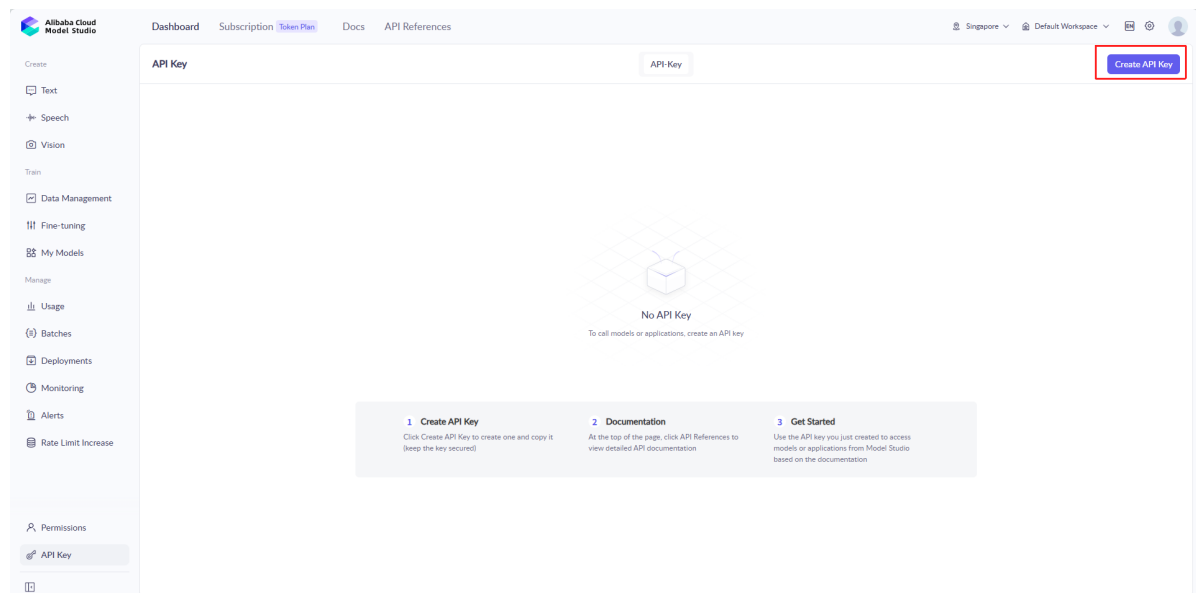
- Multiple microphone devices
- USB microphone is not the system default device
- Speaker output is directed to the wrong device

If the voice program starts but you cannot hear sound or record sound, check here first.

4.3 DashScope Authentication and ASR Model

Apply for an API key by logging into Alibaba Cloud Bailian.

Link: [Large Model Service Platform Bailian Console](#)



```
DASHSCOPE_API_KEY=  
DASHSCOPE_API_URL=https://dashscope.aliyuncs.com/api/v1/services/aigc/multimodal-  
generation/generation  
DASHSCOPE_ASR_MODEL=qwen3-asr-flash
```

Where:

- `DASHSCOPE_API_KEY`: Must fill in a valid API-KEY
- `DASHSCOPE_API_URL`: Default is already provided
- `DASHSCOPE_ASR_MODEL`: Default voice recognition model

If this is not configured correctly, common phenomena include:

- Recognition fails after program startup
- Returns authentication error
- Voice recognition request times out or reports model error

4.4 TTS Voice Broadcast Configuration

```
ENABLE_TTS=false  
WAKE_ACK_TEXT=I am here.
```

If you want voice broadcast after voice recognition, you can enable it:

```
ENABLE_TTS=true
```

Current common TTS parameters include:

- `DASHSCOPE_TTS_MODEL`
- `DASHSCOPE_TTS_VOICE`
- `DASHSCOPE_TTS_VOLUME`
- `DASHSCOPE_TTS_SPEECH_RATE`

It is recommended to get ASR working first before considering enabling TTS.

4.5 OpenClaw Gateway Configuration Reference

```
OPENCLAW_CONFIG_PATH=~/.openclaw/openclaw.json  
OPENCLAW_TIMEOUT=70 # Maximum retry time, waiting for OpenClaw response. If it's  
a long task, it is recommended to increase this time.  
OPENCLAW_RETRY=1
```

This part of the configuration determines how the voice program sends recognition results to OpenClaw.

By default:

- The program reads `openclaw.json`
- Automatically derives the local gateway port
- Automatically reads the gateway authentication token

Therefore, if voice recognition is successful but OpenClaw does not execute instructions, check here first.

4.6 Serial Port Wake-up Configuration

```
ENABLE_SERIAL_WAKE=true  
MIC_SERIAL_PORT=/dev/ttyUSB0  
MIC_SERIAL_BAUDRATE=115200  
SERIAL_WAKE_REQUIRED=true
```

If you are using the accompanying voice module, serial port wake-up mode is usually adopted.

The main functions are:

- Wait for wake-up signal normally
- Start capturing the next segment of voice upon receiving wake-up
- Return to waiting state after processing is complete

If you do not have the accompanying serial port wake-up module, you can adjust as needed:

```
ENABLE_SERIAL_WAKE=false
```

4.7 Steps to Switch Between Chinese and English

If you need to switch voice interaction from Chinese to English, it is recommended to directly modify the language parameters in `.env` and then restart the program.

Currently, `voice_to_openc1aw/.env` enables the **Domestic Version (Bailian)** by default.

Method 1: Overseas Version (iFlytek International Edition)

Main configuration items:

```
XUNFEI_ASR_LANGUAGE=en_cn  
XUNFEI_ASR_ACCENT=mandarin
```

- Chinese recognition: `XUNFEI_ASR_LANGUAGE=zh_cn`
- English recognition: `XUNFEI_ASR_LANGUAGE=en_us`
- When the recognition language is changed to English, `XUNFEI_ASR_ACCENT` is no longer needed; it can be commented out or left with its original value unused.

Example:

```
# Chinese recognition  
XUNFEI_ASR_LANGUAGE=zh_cn  
XUNFEI_ASR_ACCENT=mandarin  
  
# English recognition  
XUNFEI_ASR_LANGUAGE=en_us  
# XUNFEI_ASR_ACCENT=mandarin
```

If TTS is also enabled and you want English broadcast, you also need to change `TTS_VOICE` to an English voice you have activated; if you just want to verify English recognition first, only change the ASR language.

Method 2: Domestic Version (Bailian)

Main configuration items:

```
DASHSCOPE_LANGUAGE=zh
```

- Chinese recognition: `DASHSCOPE_LANGUAGE=zh`
- English recognition: `DASHSCOPE_LANGUAGE=en`

Example:

```
# Chinese recognition  
DASHSCOPE_LANGUAGE=zh
```

If the Bailian solution also enables TTS, it is recommended to switch to the corresponding English voice simultaneously; otherwise, you may encounter a situation where English recognition is normal, but the broadcast still has a Chinese accent.

After modification, re-execute the startup command below to take effect.

5. Starting Voice Interaction

5.1 Prepare Virtual Environment

This step is not required for the factory image. If dependencies are not yet installed, execute first:

```
cd /home/sunrise/voice_to_openclaw
python3 -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
```

5.2 Startup Method

```
cd /home/sunrise/voice_to_openclaw
source .venv/bin/activate
python -m src.main
```

If startup is successful, the program will enter a state of waiting for wake-up or waiting for voice input.

5.3 If Other Interaction Methods Need Broadcasting

- Change the broadcast parameter configuration to True

```
ENABLE_TTS=true
```

- Additionally start an independent listener, enter in terminal:

```
cd ~/voice_to_openclaw
source .venv/bin/activate
python -m src.openclaw_session_tts
```

```
(.venv) sunrise@ubuntu:~/voice_to_openclaw$ python -m src.openclaw_session_tts
[系统] OpenClaw 会话播报器已启动
[系统] 监听目录=/home/sunrise/.openclaw/agents/main/sessions
[系统] 监听渠道=feishu,tui,whatsapp
```

6. Configuration Verification Methods

6.1 Verify if Voice Program Can Start

Execute:

```
cd /home/sunrise/voice_to_openclaw
source .venv/bin/activate
python -m src.main
```

If the program starts successfully without immediate errors, the basic environment is likely normal.

6.2 Verify Recognition and Publishing Process

Say "Hello, Xiao Ya" to wake up, then try saying:

- Tell me a joke
- Display "Hello" on the OLED

If you can see OpenClaw return text, it means:

- Microphone capture is normal
- ASR is normal
- OpenClaw gateway publishing is normal

```
2026-05-13 20:00:29 [INFO] voice_to_openclaw:run:26 - voice_to_openclaw starting
[系统] voice_to_openclaw 已启动
[系统] ASR=xunfei_ws/iat/en_us | 发布=gateway_chat | TTS=开/xunfei_ws | 唤醒=串口
[系统] 串口唤醒已启用: /dev/ttyUSB0@115200
[提示] 说“清空上下文”可以重置当前会话
[唤醒] 检测到串口信号, 开始新一轮对话
[听取] 请开始说话
[听取] 正在收音
[识别] 已捕获 2820 ms 语音, 正在转写

[你] Tell me a joke.
[发送] 正在交给 OpenClaw
[完成] OpenClaw 返回 200

[OpenClaw] why did the robot go to therapy?

Because it had too many *bugs* in
its code! 🐛

...Don't worry, I'm working through my issues one debug
at a time. 😊
[信息] 回复已裁剪为播报文本: 149 -> 90 字
```

6.4 Verify TTS [Optional]

If enabled:

```
ENABLE_TTS=true
```

Then after OpenClaw returns results, you should be able to hear voice broadcast.

7. Common Issues

7.1 Program Startup Failure

Check first:

- Whether `.venv` exists
- Whether dependencies are installed
- Whether `.env` exists

7.2 No Recognition Result After Wake-up

Check first:

- Whether `ASR_ENGINE` is configured correctly
- Whether `DASHSCOPE_API_KEY` is valid
- Whether microphone device number `DEVICE` is correct

7.3 Recognition Successful but OpenClaw No Response

Check first:

- Whether `OPENCLAW_CONFIG_PATH` is correct
- Whether the gateway port in `openclaw.json` is normal
- Whether the OpenClaw gateway is started

7.4 Can Recognize but No Sound Broadcast

Check first:

- Whether `ENABLE_TTS` is `true`
- Whether `SPEAKER_DEVICE` is correct
- Whether the speaker is connected normally

7.5 Serial Port Wake-up Failure

Check first:

- Whether `ENABLE_SERIAL_WAKE` is `true`
- Whether `MIC_SERIAL_PORT` is correct
- Whether `MIC_SERIAL_BAUDRATE` is correct
- Whether the voice module serial device is recognized by the system

8. Recommended Debugging Order

It is recommended to troubleshoot the voice pipeline in the following order:

1. First confirm the `.env` file exists
2. Then confirm `DASHSCOPE_API_KEY` is filled in
3. Start the program to confirm no immediate errors
4. Check if wake-up is effective
5. Check if recognition is effective
6. Check if OpenClaw receives text
7. Finally check TTS broadcast